

# FacialX: A Robust Facial Expression Tracking System based on Multifaceted Expression Embedding

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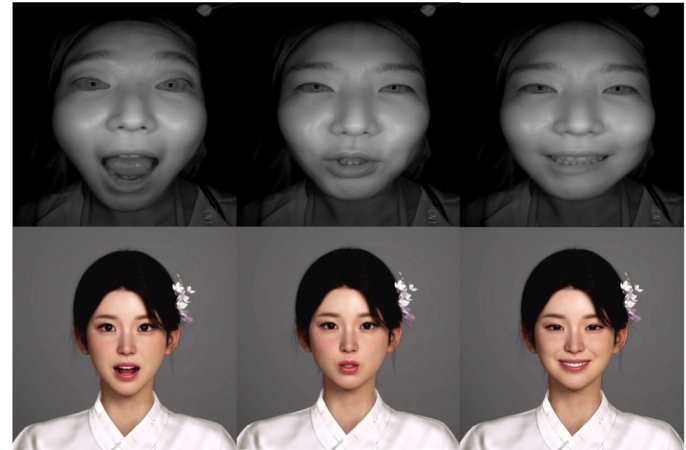
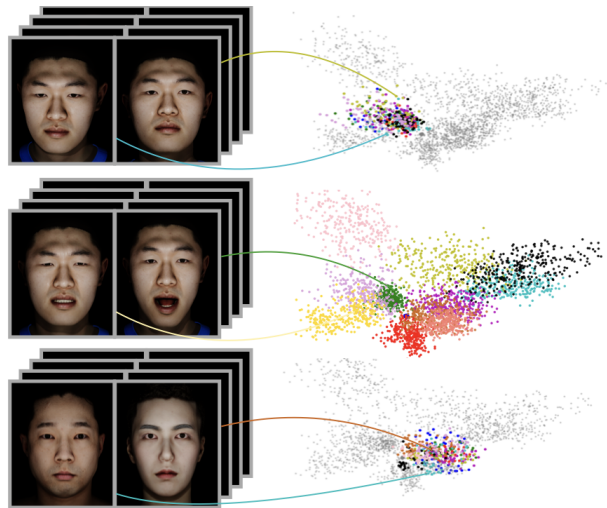
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**Figure 1: Overview of our pipeline.** We first generate extensive, yet balanced dataset that incorporate diversity in identity, camera viewpoints, and expressions to learn a multifaceted expression embedding space. Then, we apply it to facial tracking task to create a robust, high-fidelity facial expression tracking system, FacialX.

## Abstract

Recent advancements in deep learning have significantly transformed the visual effects (VFX) industry by enabling highly realistic animation of human characters through the precise capture of actors' facial expressions. Despite these innovations, many existing methodologies remain limited by specific environmental constraints, such as camera viewpoints, necessitating extensive optimization efforts to achieve optimal performance. In response to these challenges, we propose a novel approach that not only incorporates diverse facial expressions but also accounts for variations in identity and camera viewpoint. We introduce FacialX, a versatile facial tracking tool, and demonstrate its ability to generate production-quality facial animations from monocular video while consistently maintaining robust performance across a wide

range of conditions. Comparative evaluations reveal that FacialX surpasses current state-of-the-art solutions, including FACEGOOD and Apple ARKit, delivering production-quality outcomes, thereby establishing itself as a superior tool within the VFX industry.

## CCS Concepts

• **Computing methodologies** → *Motion capture*.

## Keywords

synthetic data, expression embedding, facial tracking

## ACM Reference Format:

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## 1 Introduction

The VFX industry has seen remarkable advancements in recent years, particularly through the integration of deep learning technologies for human expression analysis. Innovative works such as [Serra et al. 2022] and [Wang et al. 2024] have demonstrated excellent performance in transferring human expressions to animations. These advancements have enabled the use of head-mounted

cameras to capture actors' expressions, facilitating the realistic animation of human characters.

Despite this progress, industrial environments still face limitations with existing facial expression analysis tools. Many of these tools depend heavily on controlled environments for optimal performance, requiring consistent lighting, precise camera angles, and stable head positioning. This dependency restricts their versatility and reliability in varied production scenarios. As a result, significant time and resources are required to configure these conditions, hindering the efficiency and scalability of VFX workflows.

To address these challenges, we propose FacialX, a facial tracking tool that leverages a multifaceted embedding space. We begin by learning a comprehensive embedding space that captures variations in facial expressions, identity, and viewpoint. This embedding space is integrated into a facial tracking system designed to maintain robustness across diverse conditions. Our approach minimizes the dependency on controlled environments, enabling consistent, production-quality performance across a wide range of scenarios.

We conduct extensive evaluations to compare FacialX with existing state-of-the-art commercial solutions, including FACEGOOD [Xu et al. 2021] and Apple ARKit [Apple 2020]. Both tools are widely adopted in the industry, making them essential benchmarks for our system's evaluation. Our results demonstrate that FacialX outperforms these tools in both accuracy and robustness.

Our main contributions are

- We developed a multifaceted manifold that integrates variations in facial expressions, identity, and viewpoints.
- We integrate the learned expression embedding space into a robust facial expression tracking system, FacialX.
- We designed an evaluation metric to conduct extensive evaluations on FacialX against leading solutions to show its reliability and tracking quality.

## 2 Dataset

To effectively learn a multifaceted expression embedding space, it is essential that our dataset encompasses a diverse range of expressions, identities, and viewpoints. With advancements in digital human technology, the generation of synthetic data for human expressions has become increasingly accessible.

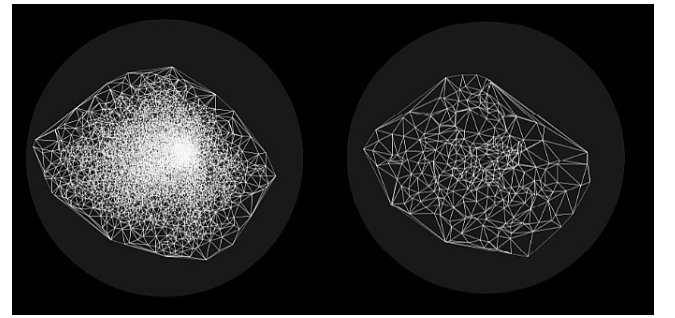
To enhance the robustness and generalization capabilities of our facial tracking tool, we collected diverse data including 66 identities and 25 distinct viewpoints. Then, we adopted the method proposed by [Park et al. 2023], which enhances the dataset by analyzing the distributional and geometric characteristics of facial embeddings and generating new, unseen expressions. However, without careful balancing, the vast number of expressions could inadvertently lead to a dataset that underrepresents diversity in identities or viewpoints, potentially compromising the model's ability to generalize across all domains.

To address this issue, we curated our dataset by balancing expressions, identities, and camera viewpoints, guided by a comprehensive quality diagnosis of the embedding space constructed in an optimal dimension. Although the use of synthetic data enabled the generation vast expressions, we identified areas of excessive density where similar or redundant samples were synthesized. To maintain

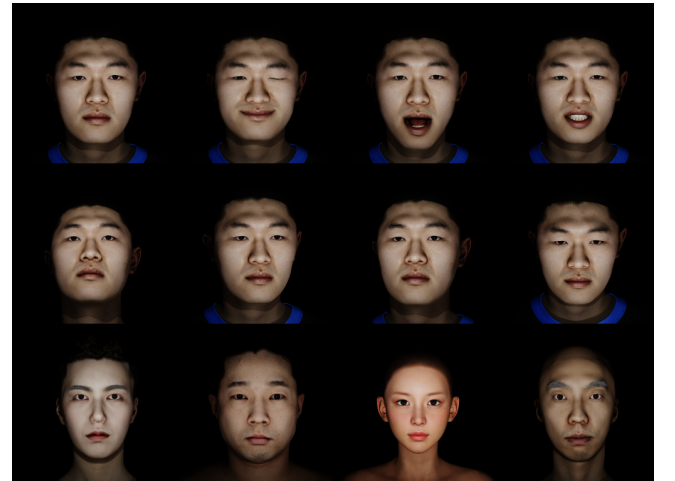
a well-balanced dataset, we implemented a data diet process, reducing the number of expressions while preserving the diversity and quality necessary for robust model training.

To demonstrate the effectiveness of our data diet, we visualized the quality of datasets using Principal Component Analysis (PCA) before and after the data reduction. (Note that the embedding dimension found is 24, then reduced for visualization.) As shown in Figure 2, while the overall distribution of the dataset was preserved, the expression density was significantly reduced. This shows that the curated dataset successfully mitigated expression bias, resulting in a more balanced and representative dataset.

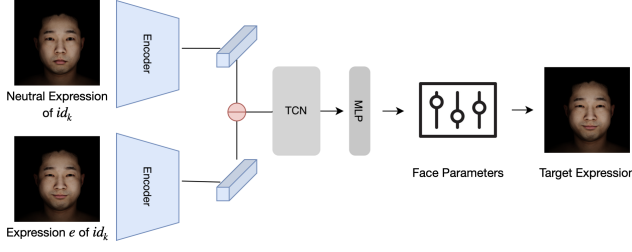
The final dataset consists of 66 identities, 25 distinct viewpoints, and 2,131 expressions. Examples of the generated dataset are displayed in Figure 3. Note that we adjusted the dataset to reflect the conditions captured through the head-mounted camera by limiting variations in lighting, viewpoints, and appearances to certain degree. This optimized dataset enables the training of an embedding space that equally represents diversity across expressions, identities, and viewpoints. Further details on data preparation can be found in the supplementary materials.



**Figure 2: Visualization of Before & After of data diet. As a result of data diet, our dataset was reduced by 78.61%.**



**Figure 3: Samples from our synthetic dataset. The first row displays diverse expressions. the second row shows diverse viewpoints, while the last row shows diverse identities.**



**Figure 4: Model architecture.** We model expressions as deviation of features to regress parameters of face model.

### 3 Training

To learn an appropriate embedding space from our expression dataset, we build on the work of [Zhang et al. 2021], which suggests that expressions should lie on a manifold relative to the neutral expression. As illustrated in Figure 4, we model facial expressions as deviations from the neutral expression, thereby explicitly disentangling the expressions.

For the encoder, we utilize a pre-trained FaceMesh network [Kartynnik et al. 2019], retraining its parameters using our dataset to take advantage of a strong initial model. By applying contrastive learning and leveraging our diverse dataset, our method effectively captures the intricate features of facial expressions across different identities and viewpoints. Following this, multilayer perceptrons (MLPs) are used to output the face model parameters, enabling the reconstruction of the input data.

During inference, we apply a one Euro filter to smooth the expression output, resulting in more natural animation.

## 4 Experiments and Results

We evaluate our method on three key fronts. First, we demonstrate that our embedding space effectively captures facial expressions. Next, we show that FacialX can track these expressions with high accuracy. Finally, we prove that FacialX maintains this accuracy across varying capture conditions.

### 4.1 Embedding Space Evaluation

Our pipeline leverages the learned embedding space to robustly track facial expressions. Therefore, we first analyze the embedding space to ensure it effectively captures the nuances of facial expressions from various angles and identities.

**Table 1: Precision@ $k$  evaluation**

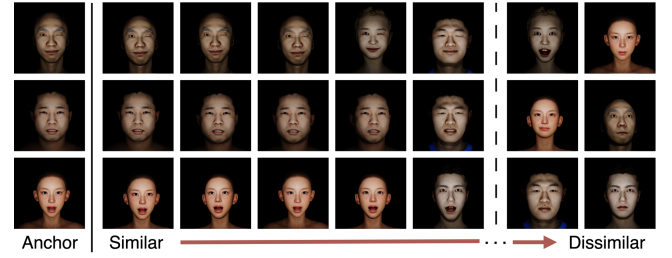
|                | $k = 3$ | $k = 5$ | $k = 10$ |
|----------------|---------|---------|----------|
| Precision@ $k$ | 95.12   | 92.73   | 90.12    |

To evaluate the quality of our embedding space, we conducted both quantitative and qualitative assessments. Quantitatively, we measured Precision@ $K$  to demonstrate that our embedding space successfully encodes facial expression features so that similar expressions are closely located in the embedding space regardless of viewpoints or identities. To measure Precision@ $K$ , we categorize expression dataset into discrete expression labels, query  $k$  nearest samples, and calculate proportion of relevant items among the top

$k$  results. Table 1. shows high Precision@ $K$  values for  $k = 3, 5, 10$  indicating that we effectively capture expressions features.

This result is also illustrated in Figure 5. where expressions closest to the anchor appear very similar while expression farthest from the anchor appear distinct. It is worth noting that expressions closest to the queried samples appear in multiple viewpoints and multiple identities indicating that our encoder learns to encode expression features disentangled from identities and viewpoints.

Overall, these evaluations confirm that our embedding space accurately represents expressions across multiple viewpoints and identities, validating its effectiveness for robust expression tracking. Further details are included in the supplementary materials.



**Figure 5: Closest and farthest expressions from each query samples.**

### 4.2 Tracking Accuracy Evaluation

To evaluate the tracking accuracy of FacialX, we compared it with two state-of-the-art products for animating EpicGames' Metahumans: Apple ARKit and FACEGOOD. However, comparing results from different facial tracking systems presents several challenges. While all these products are capable of outputting model parameters, a direct comparison of these parameters does not fully capture how accurately each system tracks human facial expressions. Additionally, there is no ground-truth value for these model parameters to serve as a definitive benchmark.

Therefore, we employed two key evaluation metrics to assess digital human animation performance. First, we developed the Frechet Expression Distance (FED), inspired by FGD [Yoon et al. 2020], to measure the realism of rendered expressions by utilizing a facial keypoint detection model [Bazarevsky et al. 2019] to measure Frechet Distance between input images and rendered scenes. We claim that utilizing facial keypoint detection model instead of InceptionV3, makes FED more suitable to evaluates the overall quality of expression tracking.

In addition to FED, we assessed tracking precision on a frame-by-frame basis by comparing pixel-level positions of facial landmarks between input video frames and rendered scenes. Specifically, we calculated the Euclidean distance between corresponding landmarks in the input and rendered images to provide a detailed measure of tracking accuracy.

For a fair comparison, we evaluated our system using data captured by each of the competing products and compared the results accordingly. We first assessed our method on data obtained using iPhone, ARKit's capturing system, and compared it with ARKit's results on the same data. We then performed a similar comparison using data captured by FACEGOOD's P1 camera. This one-to-one



**Table 2: Tracking results comparison**

| Inputs         | Data <sub>ARKit</sub> <sup>*</sup> |                | Data <sub>FACEGOOD</sub> <sup>†</sup> |               | Data <sub>FacialX</sub> <sup>‡</sup> |
|----------------|------------------------------------|----------------|---------------------------------------|---------------|--------------------------------------|
| Tracking Tools | ARKit                              | FacialX        | FACEGOOD                              | FacialX       | FacialX                              |
| FED ↓          | 87.956                             | <b>81.7902</b> | 58.248                                | <b>54.029</b> | <b>51.494</b>                        |
| FLE ↓          | 6.750                              | <b>6.061</b>   | 6.9571                                | <b>6.145</b>  | <b>5.881</b>                         |

<sup>\*</sup> Inputs from iPhone, <sup>†</sup> Inputs from FACEGOOD's P1 camera, <sup>‡</sup> Inputs from FacialX

**Figure 6: Qualitative results of FacialX**

comparison was crucial, as both ARKit and FACEGOOD are unable to predict on data captured by the other system. Visualization of our comparison method is included in the supplementary material. Finally, we evaluate FacialX on captured data using our system to demonstrate how accurately FacialX can track facial expressions.

The results of these comparative experiments are summarized in Table 2. Our findings show that FacialX outperforms both ARKit and FACEGOOD in the FED and Facial Landmark Error (FLE) metrics. Notably, FacialX achieved these results using input data from other systems, underscoring its accuracy and robustness.

Furthermore, we present qualitative results to showcase the tracking accuracy of FacialX. Visual assessments reveal that FacialX consistently maintains high fidelity in facial animation. Additional qualitative results are provided in the supplementary materials.

### 4.3 Robustness Evaluation

Finally, we demonstrate that FacialX maintains high accuracy across diverse settings, which is crucial for model's reliability in production environments. To evaluate model's robustness, we capture multiple instances of identical facial expressions from various angles and compare the variance of the resulting expression parameters.

For extensive experiments, we measured variance for two sets of expressions: basic and extraneous. Basic expressions include those commonly found in most expression datasets, such as smiling, frowning, and screaming. In contrast, extraneous expressions are less common in standard datasets and include unique movements like biting the upper lip or pressing the lips while eyes are closed.

Table 3 shows that FacialX consistently produces lower variance across different viewpoints compared to FACEGOOD. While ARKit demonstrates lower variance for basic expressions, FacialX significantly outperforms ARKit by exhibiting much smaller variance for extraneous expressions. This suggests that FacialX not only

**Table 3: Robustness evaluation**

| Methods               | ARKit         | FACEGOOD | FacialX (ours) |
|-----------------------|---------------|----------|----------------|
| Expr <sub>basic</sub> | <b>0.0006</b> | 0.0016   | 0.0012         |
| Expr <sub>extra</sub> | 0.0021        | 0.4144   | <b>0.0013</b>  |
| Mean Variance         | 0.0015        | 0.3372   | <b>0.0013</b>  |

effectively learns and generalizes a wide range of expressions but also excels in tracking less conventional expressions with high reliability. Consequently, FacialX proves to be a more dependable and versatile choice for industrial applications. Further details of the experiment, along with visual results, are included in the supplementary material.

## 5 Conclusion

In this paper, we introduced FacialX, a robust facial tracking tool designed to leverage a comprehensive embedding space that captures a wide range of facial expressions, identities, and viewpoints.

Through rigorous evaluation, we demonstrated that FacialX outperforms state-of-the-art solutions, including Apple ARKit and FACEGOOD, in both quantitative and qualitative metrics. Our method excels in tracking accuracy, as shown by superior performance in the Frechet Expression Distance (FED) and Facial Landmark Error (FLE) metrics. Additionally, FacialX exhibits exceptional robustness, maintaining low variance when tracking both basic and extraneous expressions across different viewpoints.

However, there remains room for improvement. Future efforts will focus on refining the model to fully disentangle facial expressions from identities, which is crucial for achieving natural expressions. We aim to develop a more intricate embedding capable of capturing subtle expressions and conduct more thorough experiments, including ablation studies on model architecture, comparisons with other baseline models, and an analysis of data diet. We hope to present these works in future work as a full academic paper.

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